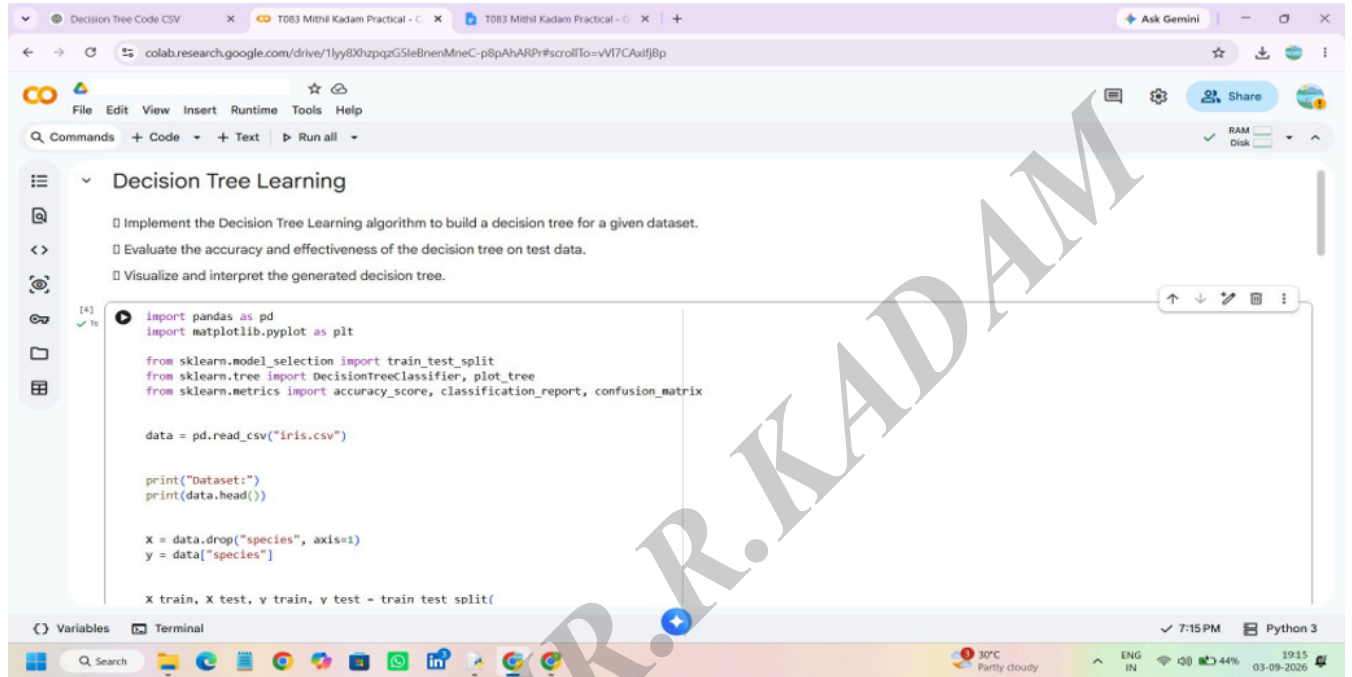


Aim:- Decision Tree Learning

Implement the Decision Tree Learning algorithm to build a decision tree for a given dataset.

Evaluate the accuracy and effectiveness of the decision tree on test data. Visualize and interpret the generated decision tree.



```
[4]: import pandas as pd
import matplotlib.pyplot as plt

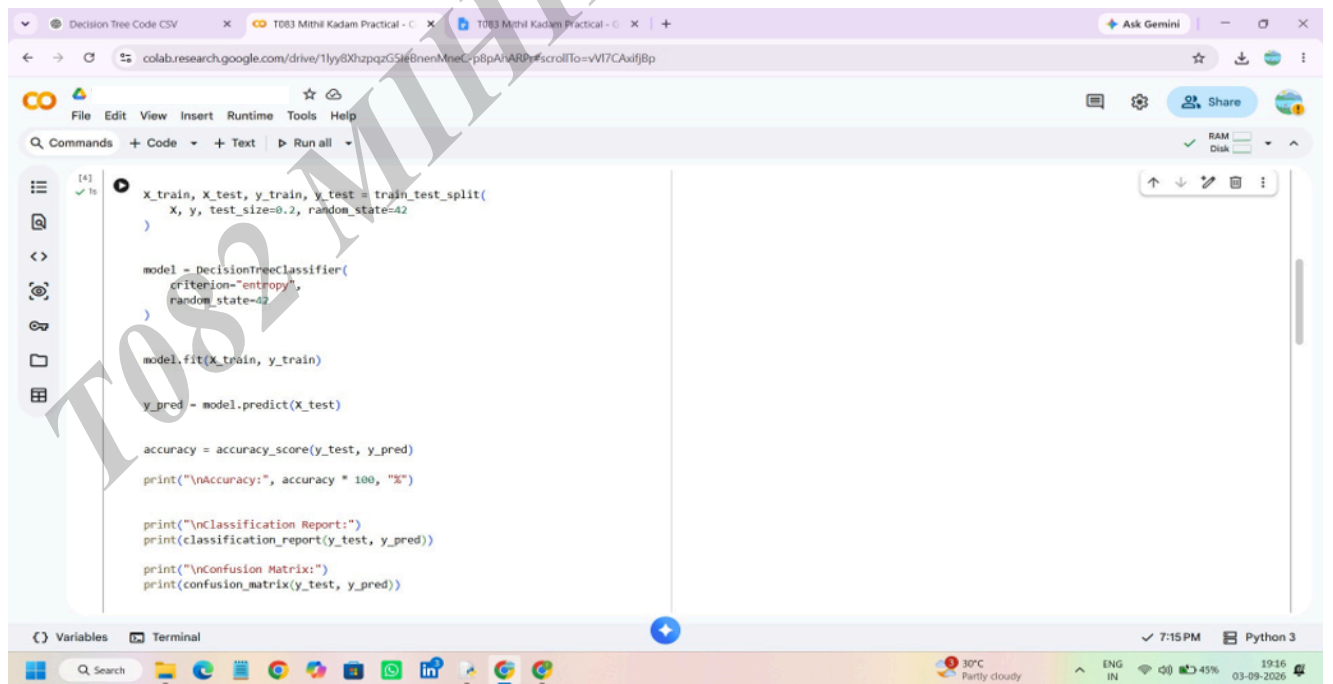
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

data = pd.read_csv("Iris.csv")

print("Dataset:")
print(data.head())

X = data.drop("species", axis=1)
y = data["species"]

X_train, X_test, y_train, y_test = train_test_split(
```



```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = DecisionTreeClassifier(
    criterion="entropy",
    random_state=42
)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", accuracy * 100, "%")

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

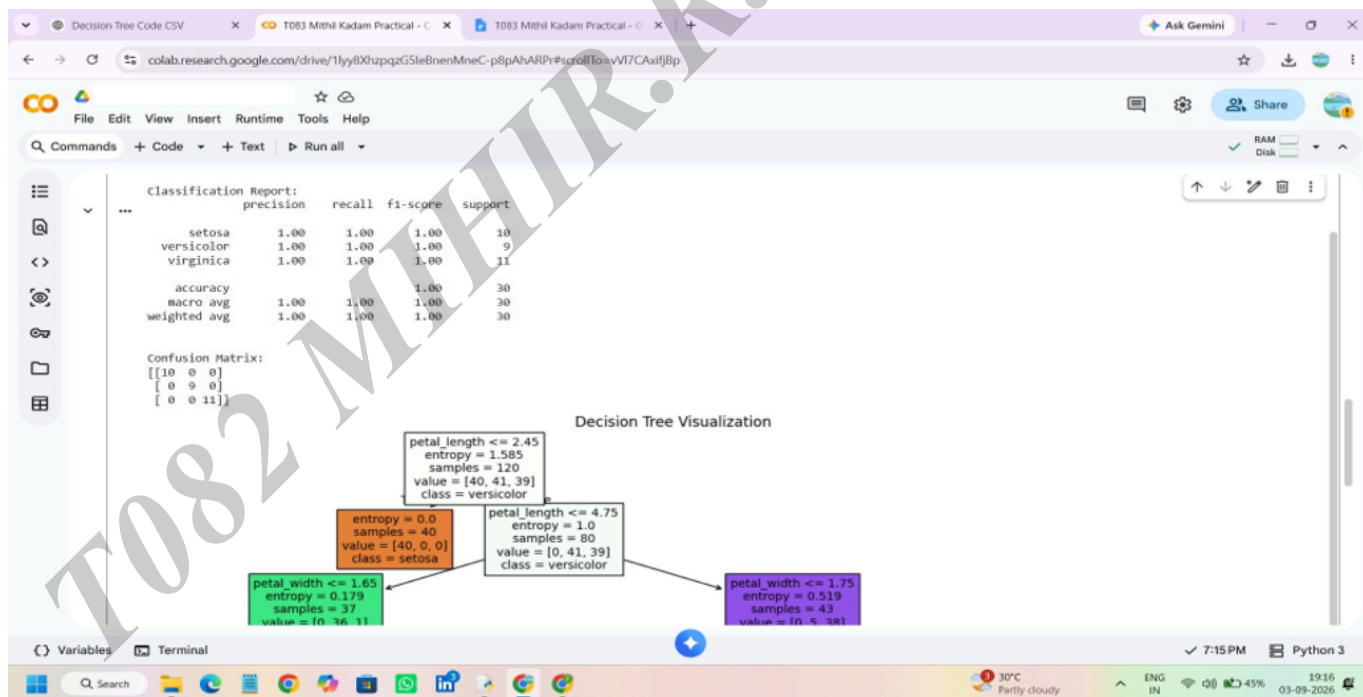
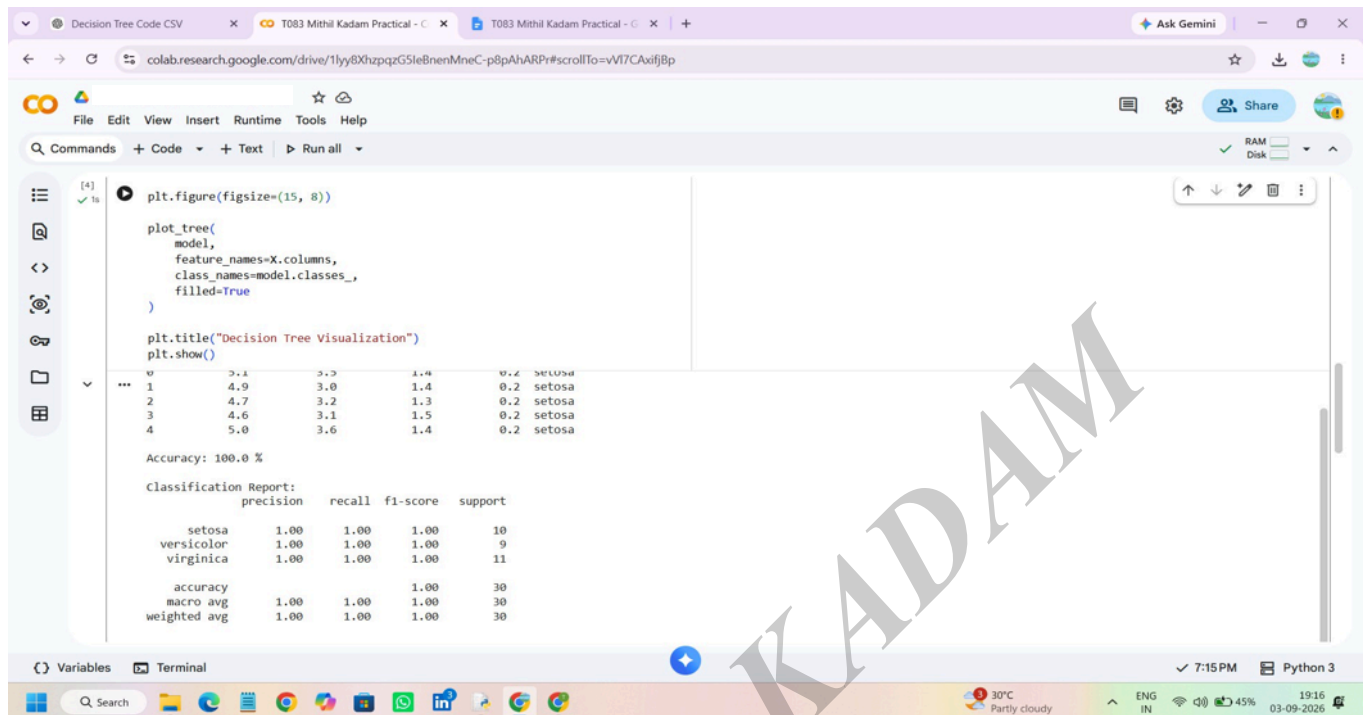
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

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